

CSF3123 - DATABASE

UNIVERSITI MALAYSIA TERENGGANU (UMT)

TITLE: LOST AND FOUND ITEM MANAGEMENT SYSTEM

GROUP 7 MEMBERS:

NO.	MATRIC NO.	NAME
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LECTURER:

Table of Content

No.	Content	Page
1.0	Introduction	
2.0	Problem Statement	
3.0	Project Scope 3.1 Users 3.2 System Scope	
4.0	Project Description	
5.0	Database Environment 5.1 Client Profile 5.2 Product/Information Flow Diagram 5.3 User Profile	
6.0	Goals and Objectives 6.1 Client Goals 6.2 User Requirement 6.3 Database Objectives	
7.0	Database Profile 7.1 Business Rules 7.2 Data Model 7.3 Normalization Process 7.4 User Interaction with the System	
8.0	Database Assessment 8.1 Validation 8.2 Weakness and Limitations 8.3 Future Enhancements	
9.0	Conclusion	

1.0 Background

The loss and recovery of personal belongings is a common issue in institutional environments such as universities. Students, staff, and visitors frequently misplace items such as student identification cards, mobile phones, wallets, stationery, and other personal belongings due to carelessness, frequent movement, and shared use of campus facilities. These incidents not only cause inconvenience to the owners but also create challenges for the management in tracking and returning lost items.

The large campus area and high student population at Universiti Malaysia Terengganu (UMT) contribute to the frequent occurrence of lost and found incidents. Currently, the management of lost and found items at UMT is mostly handled through manual methods, including verbal reports to security guards, notice boards, or information shared via social media platforms. These methods are unstructured and do not guarantee accuracy, consistency, or proper record-keeping.

The absence of a centralized system makes it difficult to efficiently store, track, and update lost and found item information. Therefore, the development of a database-based Lost and Found Item Management System specifically for UMT is essential to ensure that all reports are systematically recorded, easily accessed, and effectively managed to support the timely recovery and return of lost items.

2.0 Problem Statement

The management of lost and found items at the university level is a serious and growing concern. Students and staff frequently lose personal belongings due to factors such as carelessness, busy schedules, and frequent movement across campus. At UMT, the wide campus layout and high level of daily activity contribute to an increased number of lost and found cases.

Currently, the reporting and management of lost and found items are conducted using informal and manual approaches. These include verbal reporting to security personnel, handwritten notices, and online posts on social media. Such methods often result in incomplete records, duplicated information, and difficulty in tracking the status of items.

(a)

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(e)

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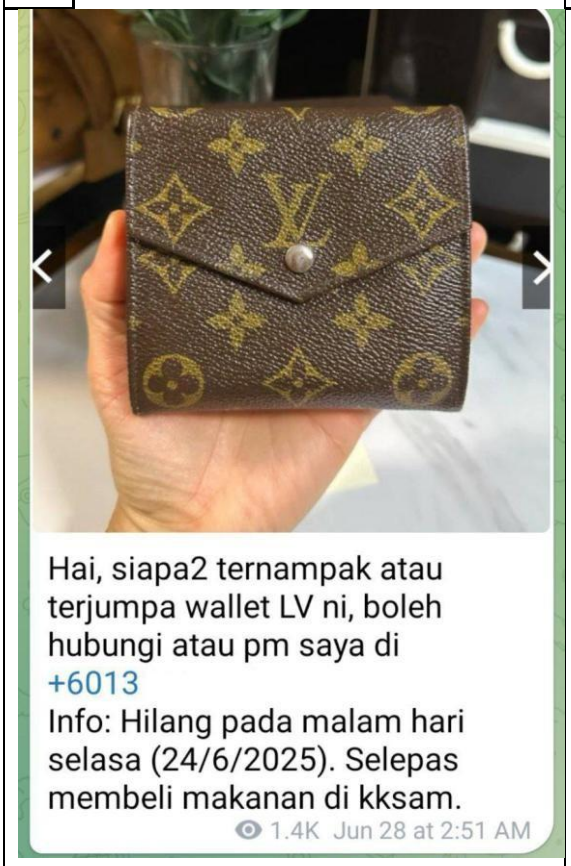
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(g)



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Figure 2.1(a)–(g) shows that lost and found items are handled using various manual and informal methods.

Furthermore, there is no centralized database that records detailed information about lost and found items, including their current status and ownership. This lack of structured data makes it difficult for university management and security personnel to monitor item recovery and ensure that items are returned to their rightful owners. Without a proper system, the process remains inefficient and prone to errors, highlighting the need for a dedicated database system to manage lost and found items within the UMT campus.

3.0 Project Scope

3.1.1 Users (Target Users)

The system is designed for use within the UMT area only. The target users of the system are:

- Guard

  Acts as the administrator of the system.
- Finder

  Students or staff who report items that they have found within the UMT campus.
- Owner

  Students or staff who report lost items and claim found items.

3.1.2 System Scope

The scope of the Lost and Found Item Management System includes:

- Reporting lost items by owners within the UMT campus.
- Reporting found items by finders within the UMT campus.
- Storing information about items, finders, owners, and guards in a database.
- Approving and updating item records by the guard.
- Submitting and processing claims by owners, verified by the guard.
- Tracking item status through its lifecycle: Pending, Lost, Found, Claimed.

4.0 Data Description

The Lost and Found Item Management System is developed to address the inefficiencies of managing lost and found items using manual methods at UMT. Manual record-keeping systems often result in unorganized records, missing information, data duplication, and difficulty in tracking item status, which can reduce management efficiency and data reliability (Connolly & Begg, 2015). These issues are common in environments with a high volume of users, such as university campuses, where lost items are frequently reported.

A centralized database system is proposed to overcome these problems by systematically storing and managing lost and found item information, including item details, finder information, owner information, and item status. Centralized databases improve data consistency and accuracy by maintaining all records in a single structured repository, making information easier to access, update, and retrieve (Elmasri & Navathe, 2016). In this system, guards are responsible for verifying and updating data, which ensures controlled access and improves data reliability.

Furthermore, the database is designed using proper normalization techniques to reduce data redundancy and prevent data inconsistency. Database normalization helps organize data into well-structured relations where all non-key attributes depend on the primary key, thereby improving data integrity and overall system efficiency (Silberschatz, Korth, & Sudarshan, 2020). By implementing this database system, the management of lost and found items at UMT can become more efficient, accurate, and reliable, providing a better experience for both users and administrators.

5.0 Database Environment

This section describes the database environment for the Lost and Found Item Management System developed for UMT. It includes the client profile, the information flow within the system, and the profiles of users who interact with the database.

5.1 Client Profile

UMT is a public university located in Kuala Nerus, Terengganu. The university has a large campus area with a high population of students, academic staff, non-academic staff, and visitors. Due to frequent daily activities such as lectures, laboratory sessions, co-curricular programs, and campus events, incidents involving lost and found personal belongings occur regularly.

Currently, the management of lost and found items at UMT is handled using manual methods such as verbal reports, notice boards, and social media postings. These approaches are unstructured and do not provide a reliable way to store, track, and update item records.

The Lost and Found Item Management System is proposed to provide a centralized database system. This database allows all lost and found item information to be recorded systematically, reducing data redundancy and improving record accuracy

UMT would benefit from the database system in several ways:

1. Provides a centralized platform to store and manage lost and found item records.
2. Improves tracking of item status such as pending, lost, found, and claimed.
3. Reduces information loss caused by manual and informal reporting methods.
4. Enhances efficiency for security guards in managing item records.
5. Supports faster and more accurate recovery of lost items by rightful owners.

The database stores essential information related to:

- Lost and found item details
- Finder information
- Owner information
- Guard records

5.2 Information Flow Diagram

A diagram of how the database implementation will fit into the station's information flow/business processes is shown in Figure 2.

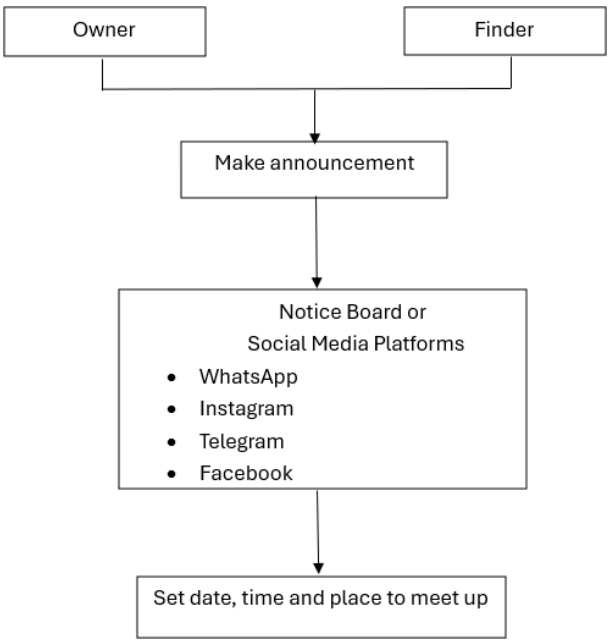
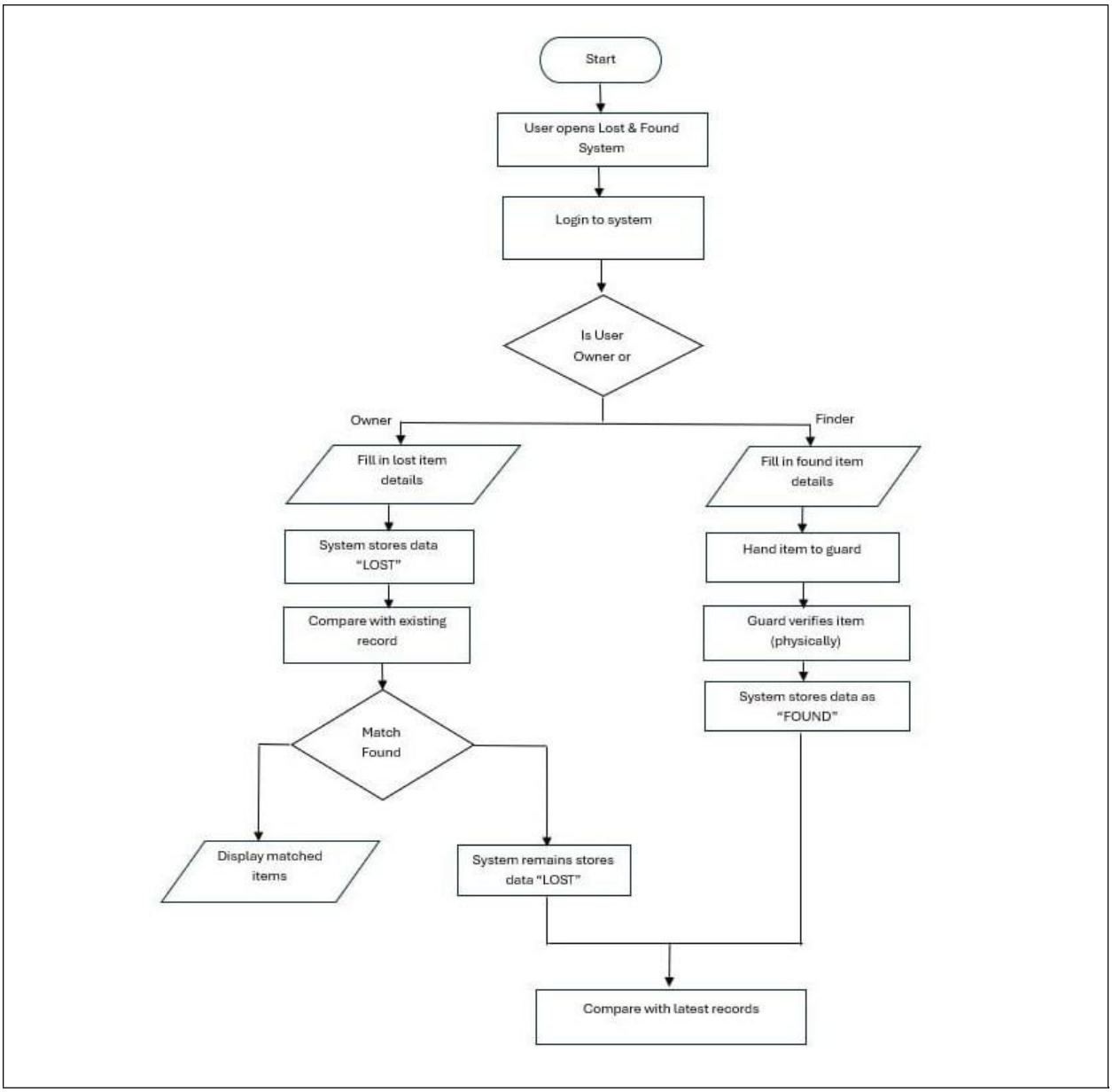


Figure 1: Information flow before database implementation


Start
User opens Lost & Found System
Login to system
Is User Owner or
Owner
Finder
Fill in lost item details
Fill in found item details
System stores data "LOST"
Hand item to guard
Compare with existing record
Guard verifies item (physically)
System stores data as "FOUND"
Match Found
Display matched items
System remains stores data "LOST"
Compare with latest records

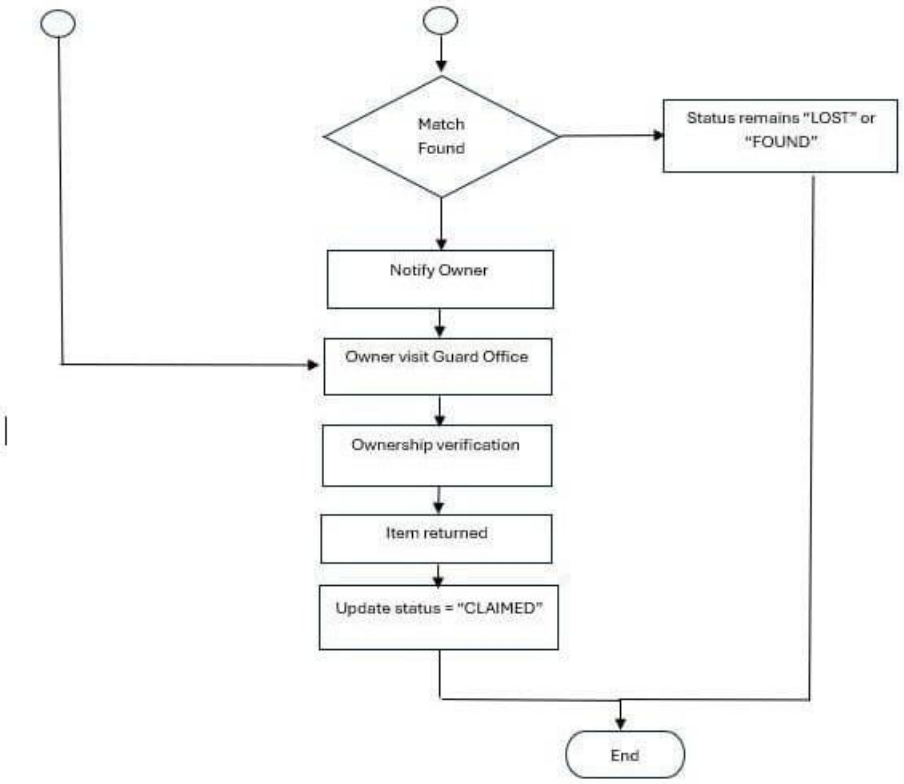


Figure 2: Information flow after database implementation

5.3 User Profile

There are three main groups of users who interact with the Lost and Found Item Management System. Each user interacts with the system differently according to their role.

i. **Guard**

Guards are university security personnel responsible for managing the Lost and Found Management System. They have full access to the database and are tasked with receiving and verifying items submitted by finders before these items are recorded in the system. After verification, guards approve and record items in the database and manage all item records, including updating their status through the lifecycle: Pending → Lost / Found → Claimed. Guards are also responsible for verifying claims submitted by owners to ensure that items are returned to the rightful owner. Additionally, they manage the information of both finders and owners in the system, ensuring the database remains accurate and up-to-

date. Guards are expected to have moderate computer experience and basic knowledge of database operations, including data entry, record management, and general system usage.

ii. **Finders**

Finders are students or staff members who discover lost items within the UMT campus. Their role is to report found items through the system and provide accurate details, including item description and their contact information. However, the item is not immediately recorded in the database; the finder must first hand over the physical item to the guard for verification. Once the guard approves and receives the item, it is then recorded in the system with the status Found. Finders do not directly access or manage the database and are expected to have basic computer literacy, as all system interactions are mediated through the guard.

iii. **Owners**

Owners are students or staff members who have lost personal belongings and wish to claim found items. They report lost items through the system, which are stored in the database with the status Lost. When a matching found item is reported and approved by the guard, the system notifies the owner of the potential match. The owner can then view the list of matching items and submit a claim for the item they identify as theirs. To complete the claim process, the owner must go to the guard for verification. After successful verification, the guard updates the item status to Claimed, completing the item recovery process. Owners are expected to have basic computer literacy and rely on the guard for recording, verifying, and retrieving information from the database.

6.0 Goals and Objectives

The primary clients and main users of the Lost and Found Item Management System are the university management at UMT and the security guards who manage lost and found items. The system also supports students, staff, and visitors who act as item owners or finders.

The main goal of the system is to provide a centralized database to record, manage, and track lost and found items within the UMT campus. The system helps administrators and guards monitor item status, verify claims, and ensure items are returned to their rightful owners efficiently. Additionally, the system serves as a central repository for compiling and generating reports related to lost and found cases.

6.1 Client Goals

- Store and access information about each lost and found item, including:
  - Item name and description
  - Location where the item was lost or found
  - Date the item was reported
  - Current item status (Pending, Lost, Found, Claimed)
  - Owner and finder information
- Update and monitor the status of each item throughout the recovery process.
- Generate summary reports to provide an overview of lost and found activities, such as:
  - Total number of lost items reported
  - Total number of found items recorded
  - Number of items successfully returned to owners
  - Number of unclaimed items
  - Lost and found cases within a specific date or location

6.2 User Requirement

- A field to indicate whether the item is reported as lost or found
- A status field to track item progress (Pending, Lost, Found, Claimed)
- Fields to store owner and finder contact information
- A date field to record when the item was lost, found, or claimed
- A description field to provide additional identifying details about the item

6.3 Database Objectives

- To design a structured and centralized database for managing lost and found items at UMT.
- To enable systematic storage and tracking of item information, including item details, finder and owner information, and item status (Pending, Lost, Found, Claimed).
- To improve the efficiency, accuracy, and consistency of lost and found item management by reducing data redundancy through proper database design and normalization.

7.0 Database Profile

This section describes the design and structure of the database used in the Lost and Found Item Management System at UMT, including business rules, data model, normalization process, and user interaction with the system. The database is designed to ensure systematic storage, accurate tracking, and easy retrieval of all lost and found item information.

7.1 Business Rules

7.1.1 Tasks Performed by the Database System

- Store information about lost and found items.
- Record item details such as name, location, date, description, and status.
- Allow Finder to report found items and submit item details.
- Allow Owner to report lost items and submit claims for matching found items.
- Allow Guard to approve, verify, and update item records.
- Track relationships between Guard, Item, Finder, and Owner.
- Update item status (Pending, Lost, Found, Claimed).
- Support basic CRUD (Create, Retrieve, Update, Delete) operations.

7.1.2 Problems Solved by the Database System

- Eliminates manual record keeping of lost and found items, reducing human error and effort.
- Reduces data redundancy and inconsistency by providing a structured storage system.
- Improves tracking of item ownership and status, making it easier to match lost and found items.
- Prevents loss of information due to informal reporting methods such as verbal reports or social media postings.
- Provides a centralized platform for managing lost and found items, allowing guards, owners, and finders to interact efficiently with the system.

7.2 Data Model

7.2.1 Entity Relationship Diagram (ERD)

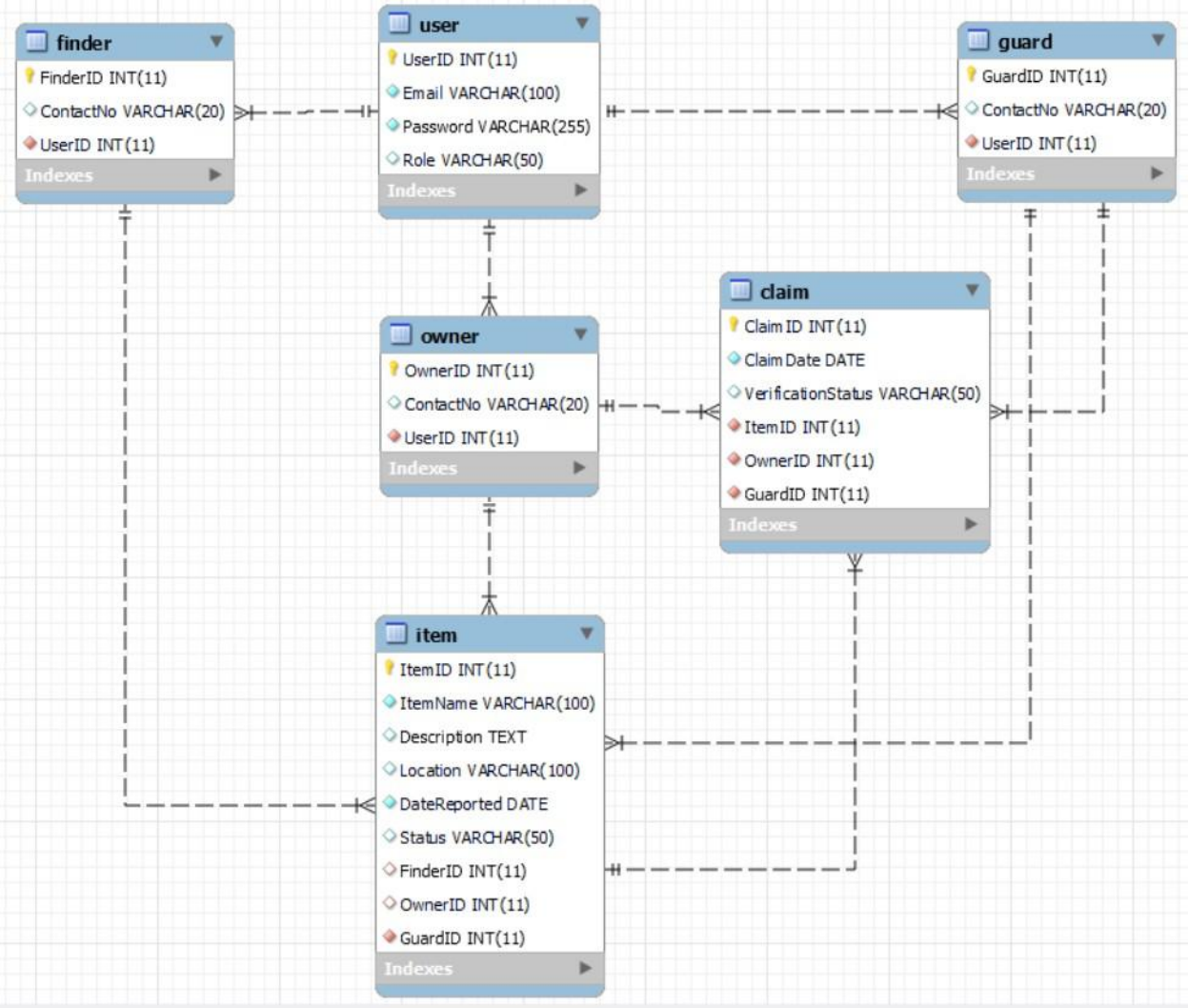


Figure 7.2.1 Entity Relationship Diagram (ERD) using Crow's Foot

7.2.2 Description of Entities, Attributes, and Relationships

i. ITEM Table

The ITEM table is the central entity in the Lost and Found Item Management System. It stores detailed information about each lost or found item reported to the system. Each item record is uniquely identified by an automatically generated Item ID (ItemID), which serves as the primary key. Other attributes stored in the ITEM table include ItemName, Description, Location, DateReported, and Status. The Status attribute indicates the current condition of the item, such as Pending, Lost, Found, or Claimed. The ITEM table is directly related to the GUARD, FINDER, and OWNER tables. It has a many-to-one relationship with the GUARD table, as each item is

approved or updated by one guard. It also has a many-to-one relationship with the FINDER table, since one finder may report multiple items. In addition, the ITEM table has a many-to-one relationship with the OWNER table, as one owner may claim more than one item. The GuardID, FinderID, and OwnerID fields function as foreign keys linking the ITEM table to the GUARD, FINDER, and OWNER tables respectively.

ii. GUARD Table

The GUARD table stores information about the guards who are responsible for managing, approving, and updating lost and found item records. Each guard is uniquely identified by a Guard ID (GuardID), which serves as the primary key. The GUARD table is related to the ITEM table in a one-to-many relationship, as one guard can approve or update multiple items. This structure allows the system to track which guard is responsible for handling each lost and found transaction, ensuring accountability and proper management of item records.

iii. FINDER Table

The FINDER table stores information about individuals who report found items to the system. Each finder is uniquely identified by a Finder ID (FinderID), which serves as the primary key. Other attributes in the FINDER table include ContactNo and UserID, which links the finder to the user account in the system. The FINDER table is connected to the ITEM table in a one-to-many relationship, as a single finder may report multiple found items. The FinderID in the ITEM table acts as a foreign key, allowing each reported item to be associated with the correct finder.

iv. OWNER Table

The OWNER table stores information about individuals who report lost items and claim found items. Each owner record is uniquely identified by an Owner ID (OwnerID), which serves as the primary key. Additional attributes include ContactNo and UserID, linking the owner to the

system's user account. The OWNER table is connected to the ITEM table in a one-to-many relationship, as one owner may claim more than one item. The OwnerID in the ITEM table serves as a foreign key, ensuring that all lost and claimed items are correctly associated with their respective owners.

v. CLAIM Table

The CLAIM table stores information about attempts by owners to claim lost items and the verification process conducted by guards. Each claim record is uniquely identified by a Claim ID (ClaimID), which serves as the primary key. Other attributes in the CLAIM table include ClaimDate, which records the date the claim was made, and VerificationStatus, which indicates whether the claim is Pending, Approved, or Rejected. The CLAIM table is linked to the ITEM, OWNER, and GUARD tables through foreign keys. The ItemID field connects the claim to the specific item being claimed, the OwnerID field links the claim to the owner who is attempting to recover the item, and the GuardID field identifies the guard responsible for verifying and approving the claim. This structure allows the system to track which owner made the claim, which item is involved, and which guard verified the claim. Each ITEM can have multiple claims, creating a one-to-many relationship between ITEM and CLAIM, while each OWNER can make multiple claims and each GUARD can verify multiple claims, maintaining accountability and accurate tracking throughout the claim and recovery process.

7.3 Normalization Process

7.3.1 UNF

At the UNF stage, all information related to lost and found items is stored in one table, causing redundancy and repetition of data.

UNF Table: Lost and Found Record

User ID	Email	Password	Role	Contact No	Finder ID	Owner ID	Guard ID	Item ID	ItemName	ItemDescription	ItemLocation	DateReported	ItemStatus	Claim ID	ClaimDate	VerificationStatus
1	finder1@umt.edu.my	f123	Finder	01210000001	1	NULL	1	1	Wallet	Black leather wallet	Library	2024-12-01	Claimed	1	NULL	NULL
2	finder2@umt.edu.my	f234	Finder	01210000002	2	NULL	2	2	Bottle	Blue water bottle	Gym	2024-12-02	Found	2	NULL	NULL
3	finder3@umt.edu.my	f345	Finder	01210000003	3	NULL	3	3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	NULL
4	finder4@umt.edu.my	f456	Finder	01210000004	4	NULL	4	4	Laptop Bag	Dell bag	Lecture Hall	2024-12-04	Found	4	NULL	NULL
5	owner1@umt.edu.my	o123	Owner	01310000001	NULL	1	5	5	Student Card	Ali Ahmad	Library	2024-12-05	Lost	5	2024-12-11	Approved
6	owner2@umt.edu.my	o234	Owner	01310000002	NULL	2	6	6	Calculator	Casio FX-570	Exam Hall	2024-12-06	Lost	6	2024-12-12	Pending
7	owner3@umt.edu.my	o345	Owner	01310000003	NULL	3	7	7	Watch	Black digital watch	Gym	2024-12-07	Lost	7	2024-12-13	Rejected
8	owner4@umt.edu.my	o456	Owner	01310000004	NULL	4	8	8	Power Bank	20000mAh	Bus Stop	2024-12-08	Lost	8	2024-12-14	Approved
9	guard1@umt.edu.my	g123	Guard	01410000001	NULL	NULL	9	9	Earphone	Wireless earbuds	Cafeteria	2024-12-09	Found	9	NULL	NULL

10	guard2@umt.edu.my	g234	Guard	01410000002	NULL	NULL	10	10	Notebook	Blue notebook	Library	2024-12-10	Found	10	NULL	NULL

-

7.3.2 1NF

At the 1NF stage, the database still uses one table only.
The difference from UNF is that all values are atomic and clearly defined.

7.3.3 2NF

In 2NF, the database is decomposed into separate tables to eliminate partial dependencies, ensuring all non-key attributes depend on the full primary key.

USER Table

UserID (PK)	Email	Password	Role
1	finder1@umt.edu.my	f123	Finder
2	finder2@umt.edu.my	f234	Finder
3	finder3@umt.edu.my	f345	Finder
4	finder4@umt.edu.my	f456	Finder
5	owner1@umt.edu.my	o123	Owner
6	owner2@umt.edu.my	o234	Owner
7	owner3@umt.edu.my	o345	Owner
8	owner4@umt.edu.my	o456	Owner
9	guard1@umt.edu.my	g123	Guard
10	guard2@umt.edu.my	g234	Guard

FINDER Table

FinderID (PK)	ContactNo	UserID
1	0121000001	1

2	0121000002	2
3	0121000003	3
4	0121000004	4

OWNER Table

OwnerID (PK)	**ContactNo**	**UserID**
1	0132000001	5
2	0132000002	6
3	0132000003	7
4	0132000004	8

GUARD Table

GuardID (PK)	ContactNo	UserID
1	0143000001	9
2	0143000002	10

ITEM Table

ItemID (PK)	ItemName	ItemDescription	ItemLocation	DateReported	Status	FinderID	OwnerID	GuardID
1	Wallet	Black leather wallet	Library	2024-12-01	Claimed	1	NULL	1
2	Bottle	Blue water bottle	Gym	2024-12-02	Found	2	NULL	1
3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	2
4	Laptop Bag	Dell bag	Lecture Hall	2024-12-04	Found	4	NULL	2
5	Student Card	Ali Ahmad	Library	2024-12-05	Lost	NULL	1	1
6	Calculator	Casio FX-570	Exam Hall	2024-12-06	Lost	NULL	2	1
7	Watch	Black digital watch	Gym	2024-12-07	Lost	NULL	3	2

8	Power Bank	20000mAh	Bus Stop	2024-12-08	Lost	NULL	4	2
9	Earphone	Wireless earbuds	Cafeteria	2024-12-09	Found	1	NULL	1
10	Notebook	Blue notebook	Library	2024-12-10	Found	2	NULL	2

CLAIM Table

ClaimID (PK)	ClaimDate	VerificationStatus	ItemID	OwnerID	GuardID
1	2024-12-11	Approved	1	1	1
2	2024-12-12	Pending	2	2	2
3	2024-12-13	Rejected	3	3	1
4	2024-12-14	Approved	4	4	2

7.3.4 3NF

In 3NF, the database is structured so that all non-key attributes depend only on the primary key, eliminating transitive dependencies.

USER Table

UserID (PK)	Email	Password	Role
1	finder1@umt.edu.my	f123	Finder
2	finder2@umt.edu.my	f234	Finder
3	finder3@umt.edu.my	f345	Finder
4	finder4@umt.edu.my	f456	Finder
5	owner1@umt.edu.my	o123	Owner
6	owner2@umt.edu.my	o234	Owner
7	owner3@umt.edu.my	o345	Owner
8	owner4@umt.edu.my	o456	Owner
9	guard1@umt.edu.my	g123	Guard

10	guard2@umt.edu.my	g234	Guard

FINDER Table

FinderID (PK)	ContactNo	UserID
1	0121000001	1
2	0121000002	2
3	0121000003	3
4	0121000004	4

OWNER Table

OwnerID (PK)	**ContactNo**	**UserID**
1	0132000001	5
2	0132000002	6
3	0132000003	7
4	0132000004	8

GUARD Table

GuardID (PK)	ContactNo	UserID
1	0143000001	9
2	0143000002	10

ITEM Table

ItemID (PK)	ItemName	ItemDescription	ItemLocation	DateReported	Status	FinderID	OwnerID	GuardID
1	Wallet	Black leather wallet	Library	2024-12-01	Claimed	1	NULL	1
2	Bottle	Blue water bottle	Gym	2024-12-02	Found	2	NULL	1
3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	2
4	Laptop Bag	Dell bag	Lecture Hall	2024-12-04	Found	4	NULL	2
5	Student Card	Ali Ahmad	Library	2024-12-05	Lost	NULL	1	1
6	Calculator	Casio FX-570	Exam Hall	2024-12-06	Lost	NULL	2	1
7	Watch	Black digital watch	Gym	2024-12-07	Lost	NULL	3	2
8	Power Bank	20000mAh	Bus Stop	2024-12-08	Lost	NULL	4	2
9	Earphone	Wireless earbuds	Cafeteria	2024-12-09	Found	1	NULL	1
10	Notebook	Blue notebook	Library	2024-12-10	Found	2	NULL	2

CLAIM Table

ClaimID (PK)	ClaimDate	VerificationStatus	ItemID	OwnerID	GuardID
1	2024-12-11	Approved	1	1	1
2	2024-12-12	Pending	2	2	2
3	2024-12-13	Rejected	3	3	1
4	2024-12-14	Approved	4	4	2

7.4 User Interaction with the System

7.4.1 User Interface Description



Figure 7.4.1 Login Interface

This interface allows all users, including guards, finders, and owners, to access the Lost and Found Management System using their registered email and password. Upon successful login, users are redirected to their respective dashboards based on their assigned roles. The tagline emphasizes the system's role as a centralized platform for managing lost and found items at UMT.

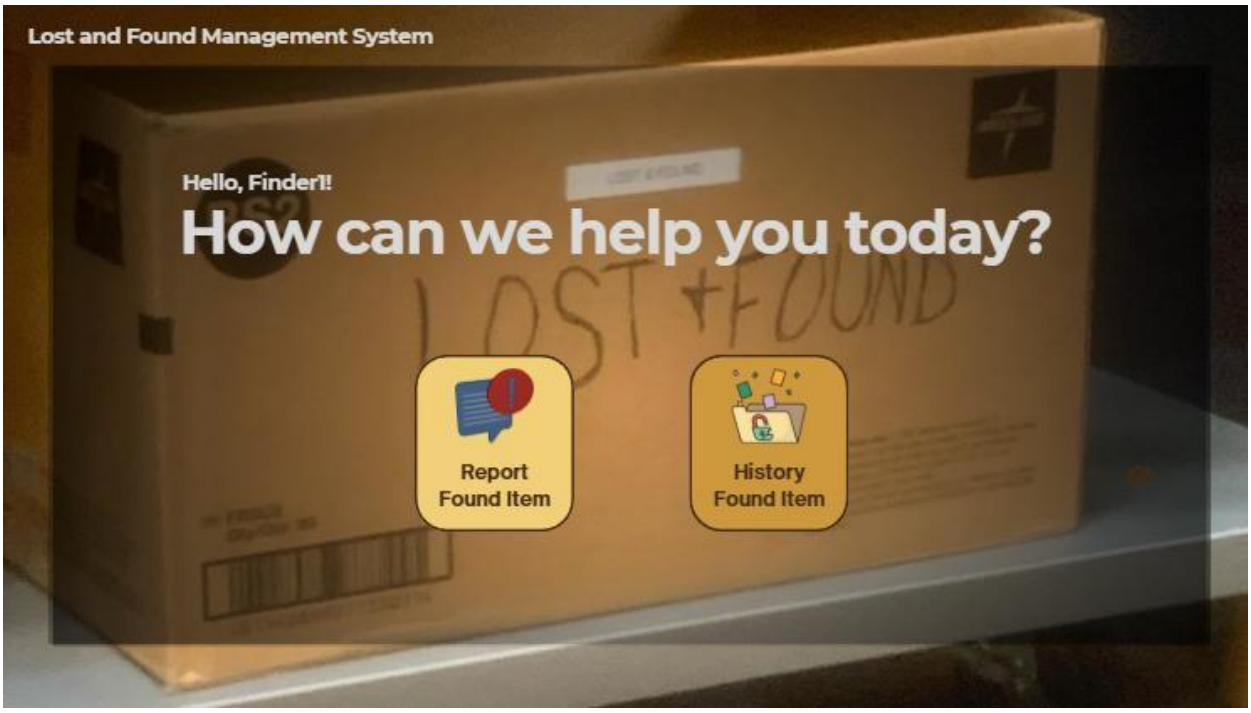


Figure 7.4.2 Finder Dashboard

The Finder dashboard provides options for users who have found items within the UMT campus. From this interface, finders can report found items and view the history of items they have reported. This interface helps ensure that found items are properly recorded and forwarded to guards for verification.



Figure 7.4.3 Report Found Item Interface

This interface allows finders to submit details of found items, including item name, description, location found, contact number, and date found. After submission, the reported item is marked as 'Pending' and will only be officially recorded in the system once the guard receives and approves the item.



Figure 7.4.4 History Found Item Interface

The History Found Item interface displays a list of items previously reported by the finder. It shows important information such as item ID, item name, description, location found, and current status (Pending or Approved). This allows finders to track the approval status of the items they reported.

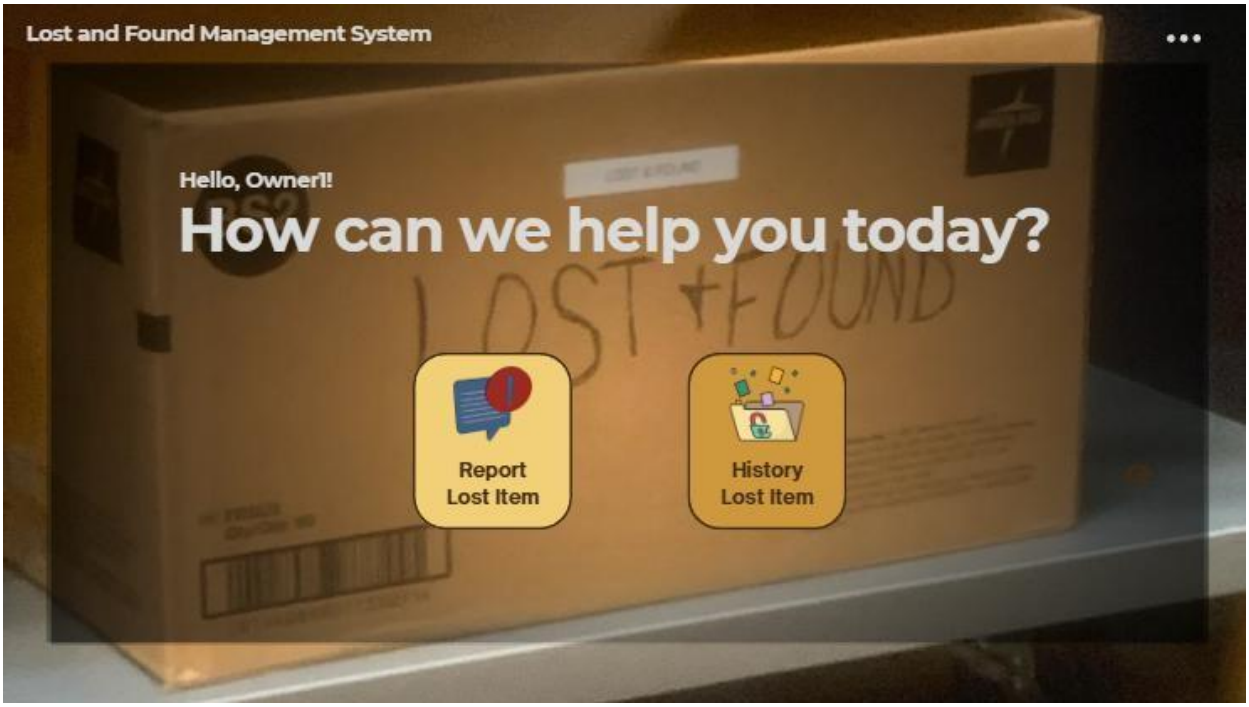


Figure 7.4.5 Owner Dashborad

The Owner dashboard provides access for users who have lost items. Owners can report lost items and view the history of their reported items. This interface enables owners to track the status of their lost items and check for potential matches with found items.



Figure 7.4.6 Report Lost Item Interface

This interface allows owners to report lost items by entering item details such as name, description, location lost, contact number, and date lost. The information is stored in the database with a 'Lost' status, allowing the system to compare it with newly reported found items.

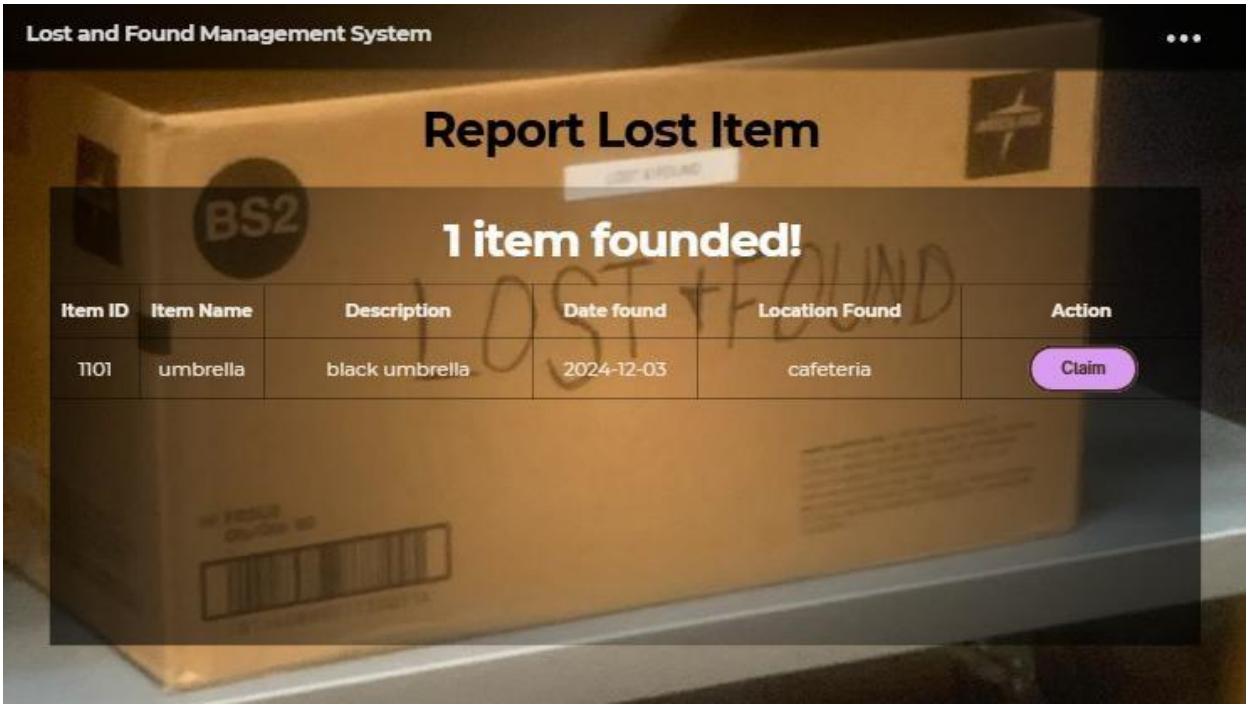


Figure 7.4.7 Match Found Interface

This interface is displayed when the system detects a matching found item for a reported lost item. Owners can view the matching item details and proceed to submit a claim request by selecting the appropriate item. The claim will then be sent to the guard for verification.

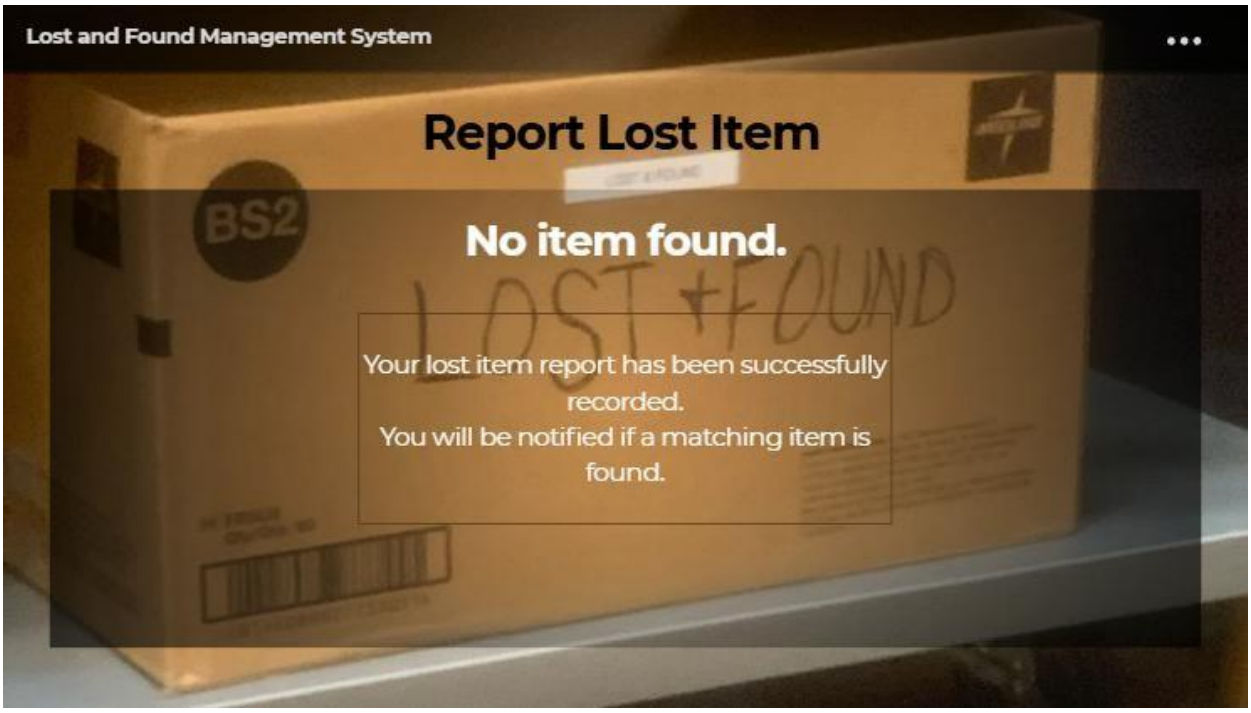


Figure 7.4.8 No Match Found Interface

This interface appears when no matching found item exists for a reported lost item. A confirmation message is shown to inform the owner that the report has been successfully recorded and that they will be notified if a matching item is found in the future.



Figure 7.4.9 History Lost Item Interface

The History Lost Item interface displays a list of lost items reported by the owner along with their statuses, such as ‘Pending’, ‘Approved’, or ‘Rejected’. This allows owners to monitor the progress of their lost item reports and claim requests.

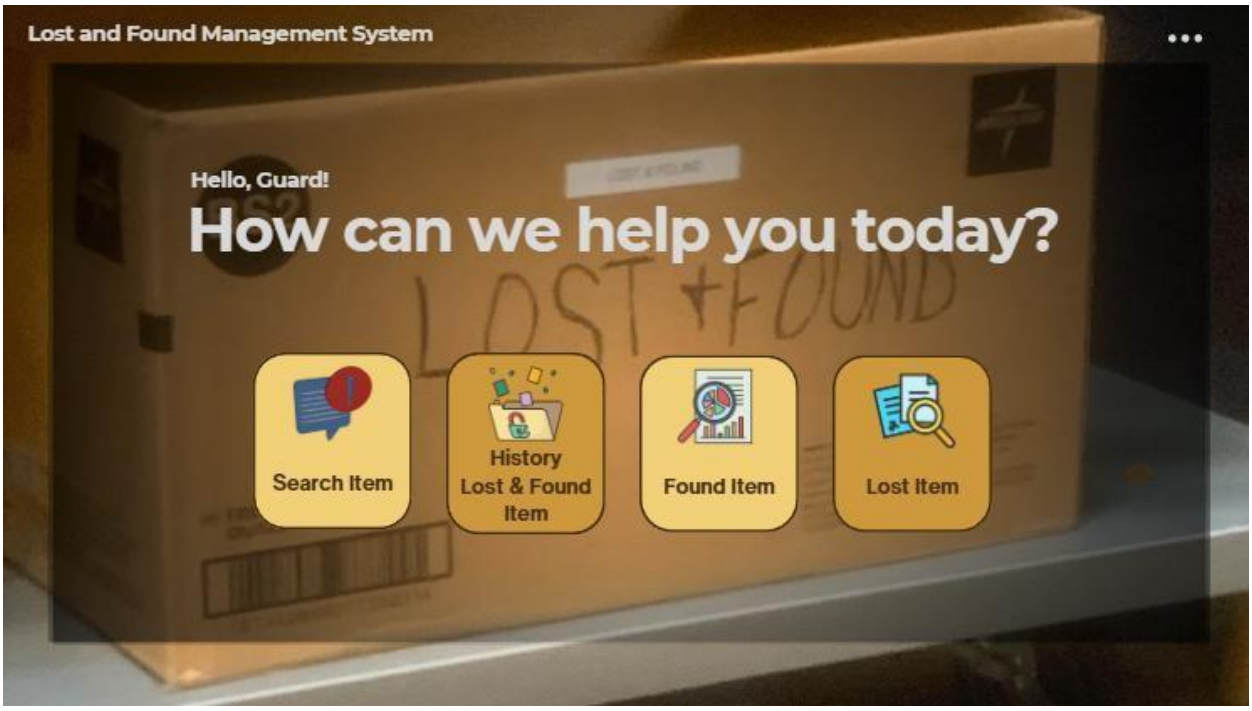


Figure 7.4.10 Guard Dashboard

The Guard dashboard serves as the main control panel for system administrators. It provides access to search items, view found items, view lost items, and review the history of lost and found records. This interface supports effective management and monitoring of all system activities.



Figure 7.4.11 History Lost & Found Interface

The History Lost & Found interface provides a complete record of all lost, found, and claimed items in the system. It allows guards to review past cases, monitor item recovery trends, and support report generation for administrative purposes.

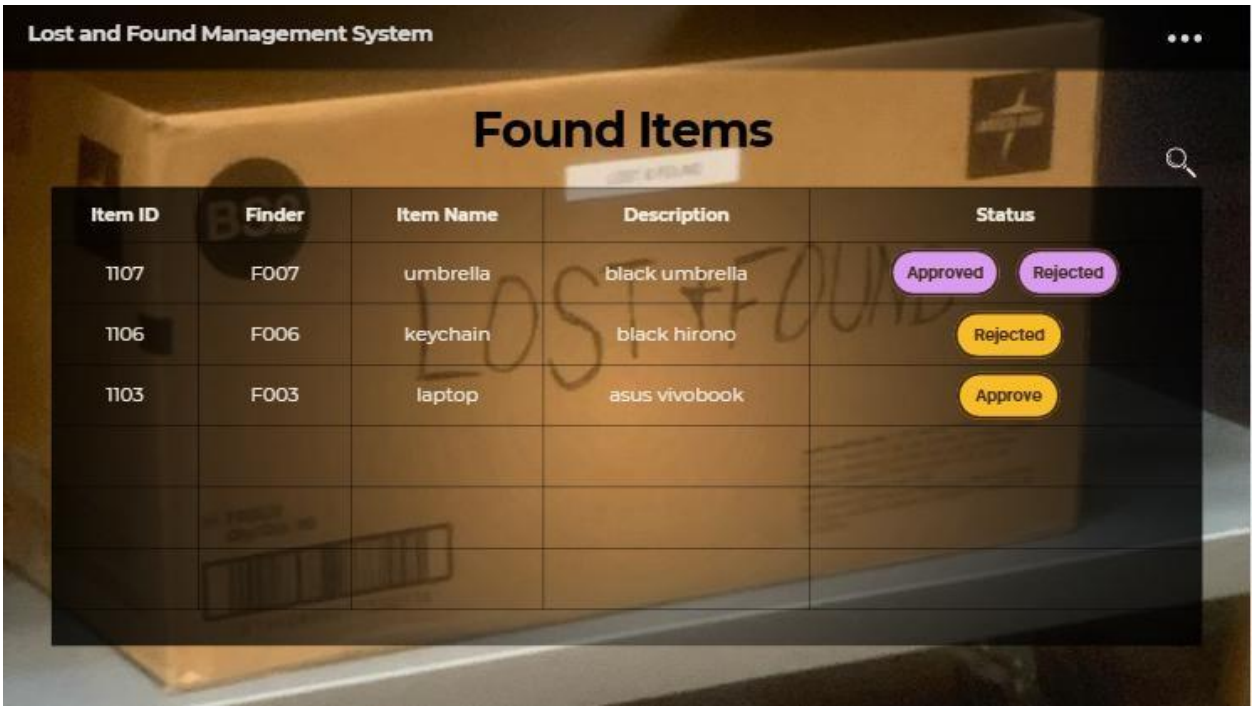


Figure 7.4.12 Found Item Interface (Guard)

The Found Item interface displays all approved found items reported by finders. Guards can review item details and approve or reject items if necessary. Approved items are made visible for owners to view and claim.



Figure 7.4.13 Lost Item Interface (Guard)

This interface displays all lost items reported by owners. Guards can monitor item status and confirm when items have been successfully claimed. Items that have been verified and returned to owners are marked with a 'Claimed' status.

7.4.3 Database Queries (CRUD)

Create Table

```sql
CREATE DATABASE assignment;
USE assignment;

CREATE TABLE User (
UserID      INT AUTO_INCREMENT,
Email       VARCHAR(100) NOT NULL UNIQUE,
Password    VARCHAR(255) NOT NULL,
Role        VARCHAR(50) CHECK (Role IN ('Finder', 'Owner', 'Guard')),
CONSTRAINT User_UserID_pk PRIMARY KEY(UserID)
);

CREATE TABLE Finder (
FinderID    INT AUTO_INCREMENT,
ContactNo   VARCHAR(20) NOT NULL,
UserID      INT NOT NULL,
CONSTRAINT Finder_FinderID_pk PRIMARY KEY(FinderID),
CONSTRAINT Finder_UserID_fk FOREIGN KEY (UserID) REFERENCES User(UserID)
);

CREATE TABLE Owner (
OwnerID     INT AUTO_INCREMENT,
ContactNo   VARCHAR(20) NOT NULL,
UserID      INT NOT NULL,
CONSTRAINT Owner_OwnerID_pk PRIMARY KEY(OwnerID),
CONSTRAINT Owner_UserID_fk FOREIGN KEY (UserID) REFERENCES User(UserID)
);
```

```sql
CREATE TABLE Guard (
GuardID     INT AUTO_INCREMENT,
ContactNo   VARCHAR(20) NOT NULL,
UserID      INT NOT NULL,
CONSTRAINT Guard_GuardID_pk PRIMARY KEY(GuardID),
CONSTRAINT Guard_UserID_fk FOREIGN KEY (UserID) REFERENCES User(UserID)
);

CREATE TABLE Item (
ItemID          INT AUTO_INCREMENT,
ItemName        VARCHAR(100) NOT NULL,
Description     VARCHAR(100),
Location        VARCHAR(100) NOT NULL,
DateReported    DATE NOT NULL,
Status          VARCHAR(50) DEFAULT 'Pending',
FinderID        INT,
OwnerID         INT,
GuardID         INT NOT NULL,

CHECK (Status IN ('Pending', 'Lost', 'Found', 'Claimed')),

CONSTRAINT Item_ItemID_pk PRIMARY KEY(ItemID),
CONSTRAINT Item_FinderID_fk FOREIGN KEY (FinderID) REFERENCES Finder(FinderID),
CONSTRAINT Item_OwnerID_fk FOREIGN KEY (OwnerID) REFERENCES Owner(OwnerID),
CONSTRAINT Item_GuardID_fk FOREIGN KEY (GuardID) REFERENCES Guard(GuardID)
);
```

```sql
CREATE TABLE Claim (
ClaimID             INT AUTO_INCREMENT,
ClaimDate           DATE NOT NULL,
VerificationStatus  VARCHAR(50) DEFAULT 'Pending',
ItemID              INT NOT NULL,
OwnerID             INT NOT NULL,
GuardID             INT NOT NULL,

CHECK (VerificationStatus IN ('Pending', 'Approved', 'Rejected')),

CONSTRAINT Claim_ClaimID_pk PRIMARY KEY(ClaimID),
CONSTRAINT Claim_ItemID_fk FOREIGN KEY (ItemID) REFERENCES Item(ItemID),
CONSTRAINT Claim_OwnerID_fk FOREIGN KEY (OwnerID) REFERENCES Owner(OwnerID),
CONSTRAINT Claim_GuardID_fk FOREIGN KEY (GuardID) REFERENCES Guard(GuardID)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	19:11:14	CREATE DATABASE assignment	1 row(s) affected	0.000 sec
3	19:11:14	USE assignment	0 row(s) affected	0.000 sec
4	19:11:14	CREATE TABLE User (UserID INT AUTO_INCREMENT, Email VARCH...	0 row(s) affected	0.031 sec
5	19:11:14	CREATE TABLE Finder (FinderID INT AUTO_INCREMENT, ContactNo ...	0 row(s) affected	0.015 sec
6	19:11:14	CREATE TABLE Owner (OwnerID INT AUTO_INCREMENT, ContactN...	0 row(s) affected	0.016 sec
7	19:11:14	CREATE TABLE Guard (GuardIDINT AUTO_INCREMENT, ContactNo ...	0 row(s) affected	0.016 sec
8	19:11:14	CREATE TABLE Item (ItemID INT AUTO_INCREMENT, ItemName VA...	0 row(s) affected	0.015 sec
9	19:11:14	CREATE TABLE Claim (ClaimID INT AUTO_INCREMENT, ClaimDateD...	0 row(s) affected	0.015 sec

Insertion

```
72  INSERT INTO User (Email, Password, Role) VALUES
73  ('finder1@umt.edu.my', 'f123', 'Finder'),
74  ('finder2@umt.edu.my', 'f234', 'Finder'),
75  ('finder3@umt.edu.my', 'f345', 'Finder'),
76  ('finder4@umt.edu.my', 'f456', 'Finder'),
77  ('owner1@umt.edu.my', 'o123', 'Owner'),
78  ('owner2@umt.edu.my', 'o234', 'Owner'),
79  ('owner3@umt.edu.my', 'o345', 'Owner'),
80  ('owner4@umt.edu.my', 'o456', 'Owner'),
81  ('guard1@umt.edu.my', 'g123', 'Guard'),
82  ('guard2@umt.edu.my', 'g234', 'Guard');
83
84  INSERT INTO Finder (ContactNo, UserID) VALUES
85  ('0121000001', 1),
86  ('0121000002', 2),
87  ('0121000003', 3),
88  ('0121000004', 4),
89  ('0121000005', 1),
90  ('0121000006', 2),
91  ('0121000007', 3),
92  ('0121000008', 4),
93  ('0121000009', 1),
94  ('0121000010', 2);
```

```
96  INSERT INTO Owner (ContactNo, UserID) VALUES
97  ('0132000001', 5),
98  ('0132000002', 6),
99  ('0132000003', 7),
100 ('0132000004', 8),
101 ('0132000005', 5),
102 ('0132000006', 6),
103 ('0132000007', 7),
104 ('0132000008', 8),
105 ('0132000009', 5),
106 ('0132000010', 6);
107
108 INSERT INTO Guard (ContactNo, UserID) VALUES
109 ('0143000001', 9),
110 ('0143000002', 10),
111 ('0143000003', 9),
112 ('0143000004', 10),
113 ('0143000005', 9),
114 ('0143000006', 10),
115 ('0143000007', 9),
116 ('0143000008', 10),
117 ('0143000009', 9),
118 ('0143000010', 10);
```

```sql
INSERT INTO Item
(ItemName, Description, Location, DateReported, Status, FinderID, OwnerID, GuardID)
VALUES
('Wallet', 'Black leather wallet', 'Library', '2024-12-01', 'Found', 1, NULL, 1),
('Bottle', 'Blue water bottle', 'Gym', '2024-12-02', 'Found', 2, NULL, 2),
('Umbrella', 'Red umbrella', 'Cafeteria', '2024-12-03', 'Found', 3, NULL, 3),
('Laptop Bag', 'Dell bag', 'Lecture Hall', '2024-12-04', 'Found', 4, NULL, 4),
('Student Card', 'Ali Ahmad', 'Library', '2024-12-05', 'Lost', NULL, 1, 5),
('Calculator', 'Casio FX-570', 'Exam Hall', '2024-12-06', 'Lost', NULL, 2, 6),
('Watch', 'Black digital watch', 'Gym', '2024-12-07', 'Lost', NULL, 3, 7),
('Power Bank', '20000mAh', 'Bus Stop', '2024-12-08', 'Lost', NULL, 4, 8),
('Earphone', 'Wireless earbuds', 'Cafeteria', '2024-12-09', 'Found', 5, NULL, 9),
('Notebook', 'Blue notebook', 'Library', '2024-12-10', 'Found', 6, NULL, 10);

INSERT INTO Claim
(ClaimDate, VerificationStatus, ItemID, OwnerID, GuardID)
VALUES
('2024-12-11', 'Approved', 1, 1, 1),
('2024-12-12', 'Pending', 2, 2, 2),
('2024-12-13', 'Rejected', 3, 3, 3),
('2024-12-14', 'Approved', 4, 4, 4),
('2024-12-15', 'Pending', 5, 5, 5),
('2024-12-16', 'Approved', 6, 6, 6),
('2024-12-17', 'Rejected', 7, 7, 7),
('2024-12-18', 'Pending', 8, 8, 8),
('2024-12-19', 'Approved', 9, 9, 9),
('2024-12-20', 'Rejected', 10, 10, 10);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	19:17:46	INSERT INTO User (Email, Password, Role) VALUES ('finder1@umt.edu....	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
11	19:17:46	INSERT INTO Finder (ContactNo, UserID) VALUES ('0121000001', 1), ('...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
12	19:17:46	INSERT INTO Owner (ContactNo, UserID) VALUES ('0132000001', 5), ('...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec
13	19:17:46	INSERT INTO Guard (ContactNo, UserID) VALUES ('0143000001', 9), ('...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
14	19:17:46	INSERT INTO Item (ItemName, Description, Location, DateReported, St...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
15	19:17:46	INSERT INTO Claim (ClaimDate, VerificationStatus, ItemID, OwnerID, Gu...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec

Retrieval

Retrieve all item details

```
148 •  SELECT * FROM Item;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ItemID	ItemName	Description	Location	DateReported	Status	FinderID	OwnerID	GuardID
1	Wallet	Black leather wallet	Library	2024-12-01	Found	1	NULL	1
2	Bottle	Blue water bottle	Gym	2024-12-02	Found	2	NULL	2
3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	3
4	Laptop Bag	Dell bag	Lecture Hall	2024-12-04	Found	4	NULL	4
5	Student Card	Ali Ahmad	Library	2024-12-05	Lost	NULL	1	5
6	Calculator	Casio FX-570	Exam Hall	2024-12-06	Lost	NULL	2	6
7	Watch	Black digital watch	Gym	2024-12-07	Lost	NULL	3	7
8	Power Bank	20000mAh	Bus Stop	2024-12-08	Lost	NULL	4	8
9	Earphone	Wireless earbuds	Cafeteria	2024-12-09	Found	5	NULL	9
10	Notebook	Blue notebook	Library	2024-12-10	Found	6	NULL	10

Item 3 ×

Output

Action Output

#	Time	Action	Message
16	19:28:30	SELECT * FROM Item LIMIT 0, 1000	10 row(s) returned

Retrieve all items that have already been found.

```
150 •  SELECT * FROM Item
151    WHERE Status = 'Found';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ItemID	ItemName	Description	Location	DateReported	Status	FinderID	OwnerID	GuardID
1	Wallet	Black leather wallet	Library	2024-12-01	Found	1	NULL	1
2	Bottle	Blue water bottle	Gym	2024-12-02	Found	2	NULL	2
3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	3
4	Laptop Bag	Dell bag	Lecture Hall	2024-12-04	Found	4	NULL	4
9	Earphone	Wireless earbuds	Cafeteria	2024-12-09	Found	5	NULL	9
10	Notebook	Blue notebook	Library	2024-12-10	Found	6	NULL	10

Item 4 ×

Output

Action Output

#	Time	Action	Message
17	19:32:15	SELECT * FROM Item WHERE Status = 'Found' LIMIT 0, 1000	6 row(s) returned

Retrieve all items by specific date

```
153 •   SELECT * FROM Item
154     WHERE DateReported = '2024-12-03';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ItemID	ItemName	Description	Location	DateReported	Status	FinderID	OwnerID	GuardID
3	Umbrella	Red umbrella	Cafeteria	2024-12-03	Found	3	NULL	3

Item 7

Output

Action Output

#	Time	Action	Message
20	19:36:26	SELECT * FROM Item WHERE DateReported = '2024-12-03' LIMIT 0, ...	1 row(s) returned

View Items with Finder Details.

```
156 •   SELECT
157         Item.ItemID,
158         Item.ItemName,
159         Finder.ContactNo
160     FROM Item
161     JOIN Finder ON Item.FinderID = Finder.FinderID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ItemID	ItemName	ContactNo
1	Wallet	0121000001
2	Bottle	0121000002
3	Umbrella	0121000003
4	Laptop Bag	0121000004
9	Earphone	0121000005
10	Notebook	0121000006

Result 10

Output

Action Output

#	Time	Action	Message
16	20:03:46	SELECT Item.ItemID, Item.ItemName, Finder.ContactNo FROM...	6 row(s) returned

Guard views claim with item and owner info.

```
163 •   SELECT
164         Claim.ClaimID,
165         Item.ItemName,
166         Owner.ContactNo,
167         Claim.VerificationStatus
168     FROM Claim
169     JOIN Item ON Claim.ItemID = Item.ItemID
170     JOIN Owner ON Claim.OwnerID = Owner.OwnerID;
171
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ClaimID	ItemName	ContactNo	VerificationStatus
1	Wallet	0132000001	Approved
2	Bottle	0132000002	Pending
3	Umbrella	0132000003	Rejected
4	Laptop Bag	0132000004	Approved
5	Student Card	0132000005	Pending
6	Calculator	0132000006	Approved
7	Watch	0132000007	Rejected
8	Power Bank	0132000008	Pending
9	Earphone	0132000009	Approved
10	Notebook	0132000010	Rejected

Result 16 ×

Output

Action Output

#	Time	Action	Message
23	20:13:15	SELECT Claim.ClaimID, Item.ItemName, Owner.ContactNo, Cl...	10 row(s) returned

Update

Guard updates item status to Claimed.

```
172 •  UPDATE Item
173    SET Status = 'Claimed'
174    WHERE ItemID = 1;
175
176 •  SELECT * FROM Item WHERE ItemID = 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ItemID	ItemName	Description	Location	DateReported	Status	FinderID	OwnerID	GuardID
1	Wallet	Black leather wallet	Library	2024-12-01	Claimed	1	NULL	1

Item 18

Output

Action Output

#	Time	Action	Message
26	20:16:49	UPDATE Item SET Status = 'Claimed' WHERE ItemID = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Guard approves a claim.

```
178 •  UPDATE Claim
179    SET VerificationStatus = 'Approved'
180    WHERE ClaimID = 2;
181
182 •  SELECT * FROM Claim WHERE ClaimID = 2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ClaimID	ClaimDate	VerificationStatus	ItemID	OwnerID	GuardID
2	2024-12-12	Approved	2	2	2

Claim 20

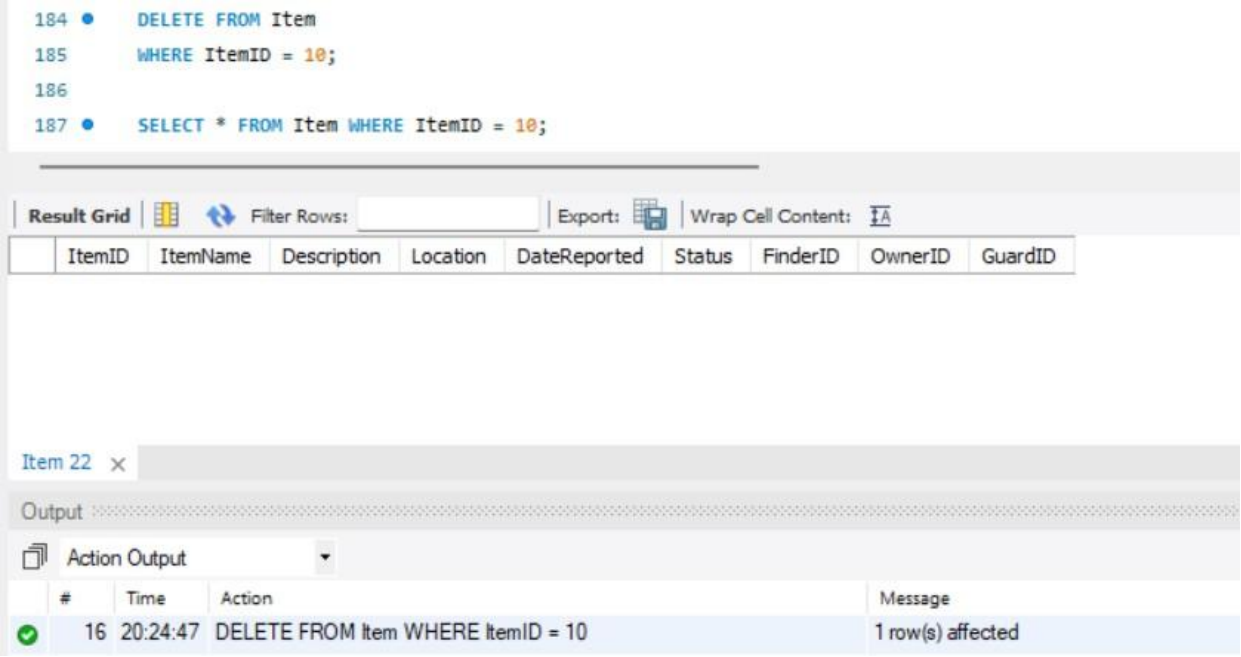
Output

Action Output

#	Time	Action	Message
29	20:20:28	UPDATE Claim SET VerificationStatus = 'Approved' WHERE ClaimID = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

<u>Deletion</u>

Guard deletes a wrong item record



8.0 Database Assessment

8.1 Validation

The database successfully fulfills the client's objectives by providing a centralized platform for managing lost and found items. All item records are stored systematically in the database and managed by the guard, ensuring accuracy and consistency of information. The system improves efficiency by replacing manual methods such as posters and verbal communication with a structured database system.

From the user perspective, the system meets its intended requirements while maintaining controlled access. Finders and owners report lost or found items directly to the guard, who then records the information into the system. This ensures that all data entered has been verified before being stored. Owners can also make claims through the guard, and item statuses such as lost, found and claimed can be tracked accurately.

In terms of database objectives, the system is well designed and normalized to reduce data redundancy and maintain data integrity. Primary keys and foreign keys are implemented to define clear relationships between the Guard, Item, Finder, and Owner entities. The database supports full CRUD operations and allows advanced searching using item attributes such as date found, category, color, description, and location, making record retrieval efficient and flexible.

9.2 Weaknesses and Limitations

Although the system meets its main objectives, several limitations exist. Access to the system is restricted to the guard, which means finders and owners must rely on manual communication to submit lost or found item information. This dependency may cause delays during busy periods. Additionally, the system does not support image uploads, which could help improve item identification and verification. The system also lacks an automatic notification feature to inform users about item status updates, and it does not include role-based login functionality for different types of users.

8.3 Potential Enhancements

To further improve the system, several enhancements can be implemented in future development. A role-based login system can be introduced to allow controlled access for finders and owners. Notification features such as email or SMS alerts can be added to inform users when an item has been found or claimed. Image upload functionality can also be implemented to improve the accuracy of item identification. In addition, reporting and analytics features can be developed to assist the guard in monitoring lost and found trends within the organization.

9.0 Conclusion

In conclusion, the Lost and Found Item Management System was developed to provide a centralized and systematic approach to managing lost and found items within an organization. The database design effectively supports the recording, updating, and tracking of item information, replacing inefficient manual methods such as posters and verbal communication.

The system successfully meets its objectives by allowing the guard to manage all item records while ensuring data accuracy and consistency. Through a well-structured and normalized database, relationships between items, finders, owners, and guards are clearly defined using primary and foreign keys. The system also supports full CRUD operations and advanced searching

based on item attributes such as date, category, color, description, and location, making data retrieval efficient.

Although the current implementation has limitations, such as restricted access for non-guard users and the absence of notification and image upload features, these limitations are acceptable for a mini project scope. Overall, the project demonstrates a solid understanding of database design concepts, including data modeling, normalization, and database management, and provides a strong foundation for future system enhancements.

10.0 References

Connolly, T., & Begg, C. (2015). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson Education.

Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson.

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). *Database system concepts* (7th ed.). McGraw-Hill.